

OS Lab06 Submission

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Task 01:

```
#include <stdio.h>
#include <string.h>
#include <sys/types.h>
#include <unistd.h>
int main(){

char Write_msg[25] = "Greetings";
char read_msg[25];
int fd[2];
pid_t pid;
pipe(fd);

if(pipe(fd) == -1){
printf("Pipe failed");
return 1;
}

pid = fork();|
if(pid < 0){
printf("Fork Failed");
return 1;
}

if(pid > 0){
close(fd[0]);
write(fd[1],Write_msg,strlen(Write_msg)+1);
close(fd[1]);
}else{
close(fd[1]);
read(fd[0],read_msg,25);
printf("Read : %s",read_msg);
close(fd[0]);
}
return 0;
}
```

Output:

```
yash-raj@yash-raj-VirtualBox:~/OS_Lab_24k-0737/Lab06$ ./T
Read : Greetingsyash-raj@yash-raj-VirtualBox:~/OS_Lab_24k-0737/Lab06$
```

Code Changes to display id's:

```
if(pid > 0){
printf("Id of Parent process is: %d\n", getpid());
close(fd[0]);
write(fd[1],Write_msg,strlen(Write_msg)+1);
close(fd[1]);
}else{
printf("Id of child is: %d\n",getpid());
printf("Id of child's parent: %d\n",getppid());
close(fd[1]);
read(fd[0],read_msg,25);
printf("Read : %s",read_msg);
close(fd[0]);
}
```

Output:

```
yash-raj@yash-raj-VirtualBox:~/OS_Lab_24k-0737/Lab06$ gcc Task01.c -o T
yash-raj@yash-raj-VirtualBox:~/OS_Lab_24k-0737/Lab06$ ./T
Id of Parent process is: 5022
Id of child is: 5023
Id of child's parent: 5022
Read : Greetingsyash-raj@yash-raj-VirtualBox:~/OS_Lab_24k-0737/Lab06$
```

Code Changes for input:

```
if(pid > 0){
printf("Id of Parent process is: %d\n", getpid());
printf("Enter Message: \n");
fgets(Write_msg, 25, stdin);
Write_msg[strcspn(Write_msg, "\n")] = 0;
close(fd[0]);
write(fd[1],Write_msg,strlen(Write_msg)+1);
close(fd[1]);

wait(NULL);
}else{
close(fd[1]);
read(fd[0],read_msg,25);
printf("Id of child is: %d\n",getpid());
printf("Id of child's parent: %d\n",getppid());
printf("Read : %s",read_msg);
close(fd[0]);
}
```

Output:

```
yash-raj@yash-raj-VirtualBox:~/OS_Lab_24k-0737/Lab06$ gcc Task01.c -o T
yash-raj@yash-raj-VirtualBox:~/OS_Lab_24k-0737/Lab06$ ./T
Id of Parent process is: 5348
Enter Message:
Hello,I am Yash
Id of child is: 5349
Id of child's parent: 5348
Read : Hello,I am Yashyash-raj@yash-raj-VirtualBox:~/OS_Lab_24k-0737/Lab06$
```

Task 02:

Producer.c

```
1 #include<stdio.h>
2 #include<stdlib.h>
3 #include<unistd.h>
4 #include<fcntl.h>
5 #include<sys/stat.h>
6 #include<string.h>
7 #define FIFO_FILE "/tmp/myfifo"
8 int main()
9 {
10     int fd;
11     char buffer[BUFSIZ];
12     ssize_t num_bytes;
13
14     mkfifo(FIFO_FILE, 0666);
15
16     fd = open(FIFO_FILE, O_WRONLY);
17
18     if(fd == -1){
19         perror("open");
20         exit(EXIT_FAILURE);
21     }
22     while(1)
23     {
24         printf("Producer: Enter a message (or 'exit' to quit): ");
25         fgets(buffer, BUFSIZ, stdin);
26
27         num_bytes = write(fd, buffer, strlen(buffer));
28
29         if(num_bytes == -1){
30             perror("write");
31             exit(EXIT_FAILURE);
32         }
33         if(strncmp(buffer, "exit", 4) == 0)
34             break;
35     }
```

Consumer.c

```
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<fcntl.h>
#include<string.h>

#define FIFO_FILE "/tmp/myfifo"

int main()
{
    int fd;
    char buffer[BUFSIZ];
    ssize_t num_bytes;

    fd = open(FIFO_FILE, O_RDONLY);

    if(fd == -1){
        perror("open");
        exit(EXIT_FAILURE);
    }
    while(1)
    {
        num_bytes = read(fd, buffer, BUFSIZ - 1);

        if(num_bytes > 0){
            buffer[num_bytes] = '\0';
            printf("Consumer: Received message: %s", buffer);
            if(strncmp(buffer, "exit", 4) == 0)
                break;
        }
    }
    close(fd);
    return 0;
}
```

Output:

```
yash-raj@yash-raj-VirtualBox:~$ gcc producer.c -o p
yash-raj@yash-raj-VirtualBox:~$ gcc consumer.c -o C
```

```
yash-raj@yash-raj-VirtualBox:~$ ./P
Producer: Enter a message (or 'exit' to quit): Hello This is Yash
Producer: Enter a message (or 'exit' to quit): hello
Producer: Enter a message (or 'exit' to quit): Lab Instructor: Sir Abdullah
Producer: Enter a message (or 'exit' to quit):
```

```
yash-raj@yash-raj-VirtualBox:~$ ./C
Consumer: Received message: Hello This is Yash
Consumer: Received message: hello
Consumer: Received message: Lab Instructor: Sir Abdullah
```

Task 03:

Producer_shm.c

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4 #include <fcntl.h>
5 #include <sys/mman.h>
6 #include <sys/stat.h>
7 #include <unistd.h>
8 int main()
9 {
10     const int SIZE = 4096;
11     const char *name = "OS";
12     const char *message_0 = "Hello";
13     const char *message_1 = "World!";
14
15     int fd;
16     char *ptr;
17
18
19     fd = shm_open(name, O_CREAT | O_RDWR, 0666);
20
21     ftruncate(fd, SIZE);
22
23     ptr = mmap(0, SIZE, PROT_WRITE, MAP_SHARED, fd, 0);
24
25     sprintf(ptr, "%s", message_0);
26     ptr += strlen(message_0);
27
28     sprintf(ptr, "%s", message_1);
29
30     printf("Producer wrote: %s %s\n", message_0, message_1);
31
32     return 0;
33 }
```

Consumer.c

```
1#include <stdio.h>
2#include <stdlib.h>
3#include <fcntl.h>
4#include <sys/mman.h>
5#include <sys/stat.h>
6#include <unistd.h>
7
8int main()
9{
10    const int SIZE = 4096;
11    const char *name = "OS";
12
13    int fd;
14    char *ptr;
15
16    // open shared memory
17    fd = shm_open(name, O_RDONLY, 0666);
18
19    // map shared memory
20    ptr = mmap(0, SIZE, PROT_READ, MAP_SHARED, fd, 0);
21
22    // read from shared memory
23    printf("Consumer read: %s\n", (char *)ptr);
24
25    // remove shared memory
26    shm_unlink(name);
27
28    return 0;
29}
```

Output:

```
yash-raj@yash-raj-VirtualBox:~$ gedit producer_shm.c
^C
yash-raj@yash-raj-VirtualBox:~$ gedit consumer_shm.c
^C
yash-raj@yash-raj-VirtualBox:~$ gcc producer_shm.c -o producer_shm -lrt
gcc consumer_shm.c -o consumer_shm -lrt
yash-raj@yash-raj-VirtualBox:~$ ^[[200~gcc producer_shm.c -o producer_shm -lrt
gcc: command not found
yash-raj@yash-raj-VirtualBox:~$ gcc producer_shm.c -o producer_shm -lrt
yash-raj@yash-raj-VirtualBox:~$ gcc consumer_shm.c -o consumer_shm -lrt
yash-raj@yash-raj-VirtualBox:~$ ./producer_shm
Producer wrote: Hello World!
yash-raj@yash-raj-VirtualBox:~$ ./consumer_shm
Consumer read: HelloWorld!
yash-raj@yash-raj-VirtualBox:~$
```


